

KHUSHI PRASHANT MEHTA

Data Scientist | Data Analyst | AI/ML Engineer

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TECHNICAL SKILLS

Programming Languages: Python, SQL, C++, JavaScript, HTML/CSS, MySQL, PostgreSQL

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, Deep Neural Networks, CNNs, Computer Vision, NLP, Logistic Regression, Recommender Systems, Time-Series Forecasting, NumPy, Pandas, Statistics & Probability, A/B Testing

Generative AI & LLMs: LLM Fine-Tuning, Retrieval-Augmented Generation (RAG), Hugging Face Transformers, LangChain, LangGraph, LlamaIndex, CrewAI, Agentic Workflows, Prompt Engineering, Tool Calling, Model Context Protocol (MCP), Vector Databases (FAISS, ChromaDB), OpenAI API, OpenRouter

Data Engineering & MLOps: ETL Pipelines, Data Processing & Cleaning, REST APIs, FastAPI, MLflow, Docker, Git, GitHub, CI/CD, AWS (EC2, S3, SageMaker), Azure ML, Google Vertex AI, Google Colab, Jupyter

Frameworks, BI & Design Tools: Flask, Streamlit, Gradio, Django, Power BI, Data Visualization, Dashboarding, ArcGIS, ArcPy, Figma, Adobe XD

PROFESSIONAL EXPERIENCE

ASML | Machine Learning Intern, Pulse Agent

May 2026 - Aug 2026

- Designed and deployed Pulse Agent, a full-stack GenAI application that lets business users ask questions in natural language, converts them to SQL on structured pulse data, and surfaces results in Power BI, replacing manual query and report requests.
- Partnered with business, logistics, and finance stakeholders to translate KPI and operational questions into Text-to-SQL.
- Fine-tuned a large language model on curated pulse datasets and SQL procedures, improving domain-specific answer accuracy and traceability on business-critical queries.
- Architected a Retrieval-Augmented Generation (RAG) pipeline that consolidated unstructured process documentation from multiple teams into a single searchable knowledge base for the agent.
- Implemented SQL metadata tagging in the natural-language interface, increasing output transparency and query traceability.
- Built time-series forecasting on 24,300+ shipment records that surfaced a 28% Q4-over-Q1 transportation cost increase, and delivered an interactive Power BI dashboard (KPIs, cost per kg, lane/mode/carrier trends) to logistics and finance leadership.

USC Marshall School of Business | Teaching Assistant, AI for Data Science

May 2025 - Present

- Reviewed and debugged Python ML codebases for 40+ students, diagnosing logic defects, unhandled edge cases, and performance bottlenecks and guiding fixes through more efficient data structures and algorithmic approaches.
- Supported development of end-to-end ML and NLP pipelines in Python (TensorFlow, scikit-learn), covering data preprocessing, feature engineering, model training and evaluation on real-world datasets.
- Mentored predictive-analytics projects from problem definition to working solution, helping students turn ambiguous business problems into well-scoped ML systems through sound model selection.
- Reinforced software engineering best practices (modular design, reusable code, systematic debugging) to move student work toward production-quality ML code.

Hashtechy | AI UI Front-End Designer

Sep 2023 - Dec 2024

- Designed and built AI-driven front-end interfaces for the Currently social media platform, integrating ML model outputs into personalized UI components and contributing to growth to 500K+ downloads and a Shark Tank India feature.
- Wrote SQL queries and Python scripts to evaluate 100+ user A/B tests, using behavioral metrics to improve retention.
- Collaborated with ML engineers to turn model-training insights into front-end features and built data visualizations for executives.

Space Application Center, ISRO | AI/ML Research Intern

Jan 2023 - Jul 2023

- Benchmarked classical ML models (k-means, Random Forest) against deep learning for land-cover classification using ArcGIS and ArcPy on multi-source satellite imagery, applying factorization to the image data.
- Designed and trained a U-Net CNN for semantic segmentation of glacier-covered land, using data augmentation, hyperparameter tuning, and fine-tuning to improve classification accuracy by 20-30%.
- Evaluated the model on 1,000+ Sentinel-2 images with IoU, Dice score, and accuracy using Python (GDAL, Rasterio, NumPy).

PROJECTS

MCP-Powered AI Assistant | Python, Model Context Protocol, Claude API, Groq, pytest

- Built an AI assistant on Anthropic's Model Context Protocol with a decoupled tool server (web search, file I/O, REST API) and LLM client, enabling Claude/Groq provider swaps with no code changes; secured tool execution via sandboxed file access.

Loan Approval Prediction | Python, Scikit-learn, Streamlit, OpenAI API

- Trained a logistic regression model (81% accuracy on 600+ applications) and deployed a Streamlit app with OpenAI-generated plain-language explanations, cutting decision time by about 40%.

AI-Powered Travel Planning Assistant | Python, Flask, OpenRouter, Mixtral

- Built a full-stack GenAI assistant that generates personalized itineraries from natural-language queries, with multi-turn memory.

EDUCATION

University of Southern California | M.S., Computer Science

Jan 2025 - Dec 2026

Indus University | B.Tech, Information Technology

Jul 2019 - Apr 2023